

Pratik Raj

+91-8409706023 | Pune, Maharashtra, India | pratikraj2000qwerty@gmail.com

github.com/PratikThisSide | linkedin.com/in/prpratikraj

pratikraj.dev | leetcode.com/u/pratikraj2000qwerty

SUMMARY

Computer Science undergraduate focused on backend and distributed systems, with hands-on experience building enterprise applications, cloud-native services, and RAG-based platforms. Skilled in Java, Spring Boot, Node.js, PostgreSQL, Redis, and AWS, with globally recognized certifications from AWS and Oracle Cloud.

EDUCATION

Bachelor of Technology in Computer Science & Engineering (CGPA: 7.28)

MIT ADT University, Pune

2023 – 2027

Class XII — Lyceum International School

2021 – 2023

Class X — Indian Public School

2020 – 2021

PERSONAL PROJECTS

ERP & Academic Intelligence Platform

Enterprise Full-Stack System | Java, Spring Boot, React.js, RAG

In Development

- Engineered a modular university ERP covering student, faculty, attendance, assessments, results, transcripts, notifications, and academic workflows using Spring Boot, React.js/TypeScript, PostgreSQL, and Redis.
- Implemented JWT-based RBAC and backend authorization with role-specific access controls, request validation, centralized exception handling, and full audit logging.
- Built deterministic academic engines and state-machine workflows for grades, SGPA/CGPA, credits, progression, eligibility, risk detection, and automated academic interventions.
- Developed versioned transcript integrity workflows using SHA-256 verification hashes, reference codes, immutable locking, supersession tracking, and public verification.
- Integrated grounded RAG with pgvector for context-aware academic assistance via semantic retrieval, keeping authoritative ERP data and deterministic business rules as the source of truth.

Campus Maintenance Scheduling System

Full-Stack Web Application

- Built a scalable platform to manage and prioritize campus maintenance requests, designed to support 1,000+ concurrent users.
- Implemented role-based access control for Admin, Staff, and User roles using Node.js and JWT for secure, session-aware authentication.
- Designed a structured relational schema in MySQL for efficient request tracking, real-time status updates, and a full audit trail.
- Tech Stack: React.js, Node.js, MySQL, JWT.

Smart Auto-Scaling URL Shortener

Distributed Cloud-Native System

- Designed a distributed URL-shortening service comparable to bit.ly, focused on high availability, low-latency redirection, and horizontal scalability.
- Built REST APIs using Node.js for URL generation, mapping, and redirection; integrated auto-scaling logic to absorb traffic spikes.
- Leveraged AWS EC2, S3, and DynamoDB/RDS for scalable compute, static asset delivery, and persistent URL storage.
- Tech Stack: Node.js, AWS EC2, S3, DynamoDB/RDS.

ACHIEVEMENTS & CERTIFICATIONS

- Maintained a 500+ day active problem-solving streak on LeetCode, demonstrating consistent algorithmic discipline.
- AWS Certified Cloud Practitioner** (Foundational) — Amazon Web Services.
- AWS Certified Solutions Architect** – Associate — Amazon Web Services.
- Oracle Cloud Infrastructure (OCI) Cloud Foundation Associate** — Oracle.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, SQL

Backend: Spring Boot, Node.js, REST APIs

Frontend: React.js, TypeScript, HTML, CSS

Databases: PostgreSQL, MySQL, Redis, DynamoDB

Cloud & DevOps: AWS (EC2, S3, RDS, DynamoDB), Docker, Git, Linux

Core Concepts: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, OOP